

STRATEGIC ANALYSIS OF CUSTOMER BOOKING & CANCELLATION BEHAVIOR

Optimizing Revenue, Occupancy, and Lead-Time Management for Resort and City Properties

DATA-DRIVEN EXECUTIVE CONSULTING REPORT

A definitive study examining 85,962 unique booking transactions across urban and resort hotel segments (2017–2019) to optimize pricing yield, mitigate cancellation leakage, and stabilize seasonal cash flow.

Prepared For:	Chief Operating Officer & Revenue Management Committee
Primary Scope:	City Hotel & Resort Hotel Operational Performance
Data Range:	July 2017 – August 2019 (119,390 Raw Observations)
Cleaned Sample:	85,962 Verified Unique Transactions
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Executive Summary

Hospitality profitability relies heavily on predicting demand patterns, pricing rooms accurately, and mitigating revenue loss from booking cancellations. This report analyzes historical customer reservation data spanning July 2017 through August 2019 from two distinct properties: an urban **City Hotel** and a leisure **Resort Hotel**.

85,962	61.4%	27.5%	\$101.30
CLEANED TRANSACTIONS	CITY HOTEL SHARE	PORTFOLIO CANC. RATE	PORTFOLIO AVG ADR

Core Executive Insights

- **Portfolio Distribution:** City Hotel accounts for 61.4% of unique bookings (N = 52,739), whereas Resort Hotel captures 38.6% (N = 33,223).
- **Demand Seasonality:** Peak monthly booking volume occurs in **August** (10,394 bookings combined), while the annual trough occurs in **January** (4,446 bookings).
- **Stay Duration Vulnerability:** Cancellation rates spike heavily on extended stays exceeding 7 nights, reaching a severe peak of **71.4%** for 10-night stays at Resort Hotel.
- **Lead Time Instability:** Lead time is the single strongest driver of cancellation risk. Reservations booked >120 days in advance exhibit cancellation rates exceeding **50–60%**.

1. Problem Statement & Dataset Overview

1.1 Strategic Problem Statement

In hospitality management, perishable room inventory represents unrecoverable revenue if unfulfilled. Unpredicted cancellations create operational challenges: overbooking risks guest displacement and brand erosion, while underbooking leads to vacant rooms and margin loss. This study establishes a data-driven framework to reduce cancellation exposure and optimize booking policies.

1.2 Cleaned Dataset Schema & Variables

Feature Name	Data Type	Business Description	Operational Impact
hotel	String	Property Identifier (City Hotel vs. Resort Hotel)	Primary segment grouping
is_canceled	Integer (0/1)	Binary indicator of booking cancellation status	Target metric for risk modeling
lead_time	Integer	Days between reservation creation & scheduled arrival	Primary predictor of cancellation
adr	Float	Average Daily Rate (Total Revenue / Paid Stay Nights)	Unit revenue performance metric
total_stay	Integer	Calculated combined stay duration (Weekend + Weekday)	Capacity planning & length stay risk

2. Data Preprocessing & Pipeline Architecture

To eliminate noise and redundant transactional logs, a rigorous 5-step data cleaning methodology was executed on the raw 119,390 observations:

- **Missing Value Imputation:** children imputed with 0; city imputed with 'Unknown'; agent & company imputed with 0.0 (indicating direct individual bookings).
- **Deduplication:** 31,994 exact duplicate rows (~26.8%) were removed to eliminate log redundancies from channel manager retries.
- **Categorical Standardization:** 'Undefined' meal categories mapped to 'No Meal' (Self-Catering).
- **Anomaly & Outlier Filtering:** Purged 168 anomalous records (1 negative ADR, 1 extreme outlier ADR at \$5,400, and 166 zero-guest administrative reservations).

3. Monthly Booking & Portfolio Demand Analysis

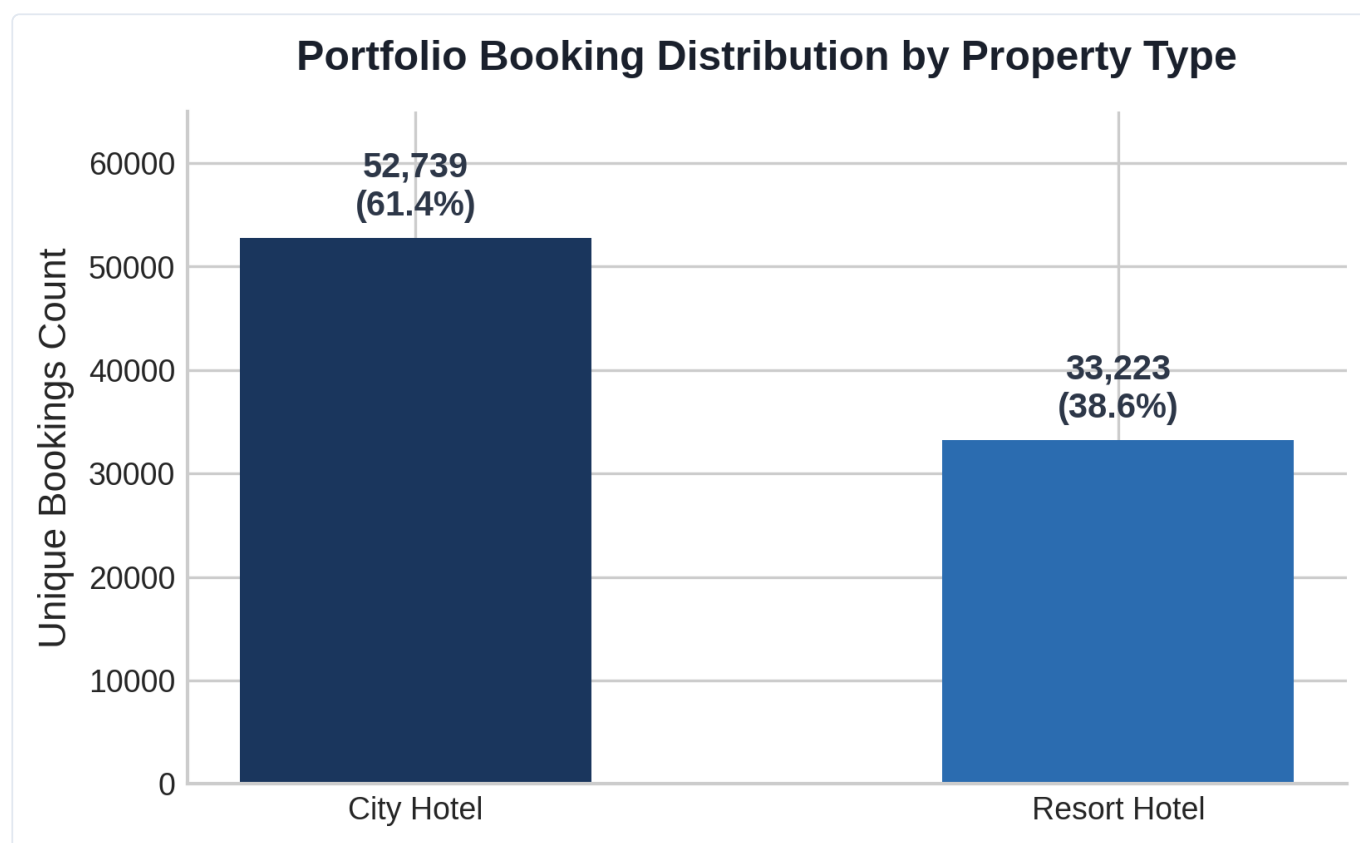


Figure 1: Unique Transaction Share between City Hotel and Resort Hotel (2017–2019)

City Hotel commands **61.35%** of the total booking volume, driven by high-velocity urban transient demand and corporate business travel. Resort Hotel accounts for **38.65%**, reflecting longer leisure stay cycles.

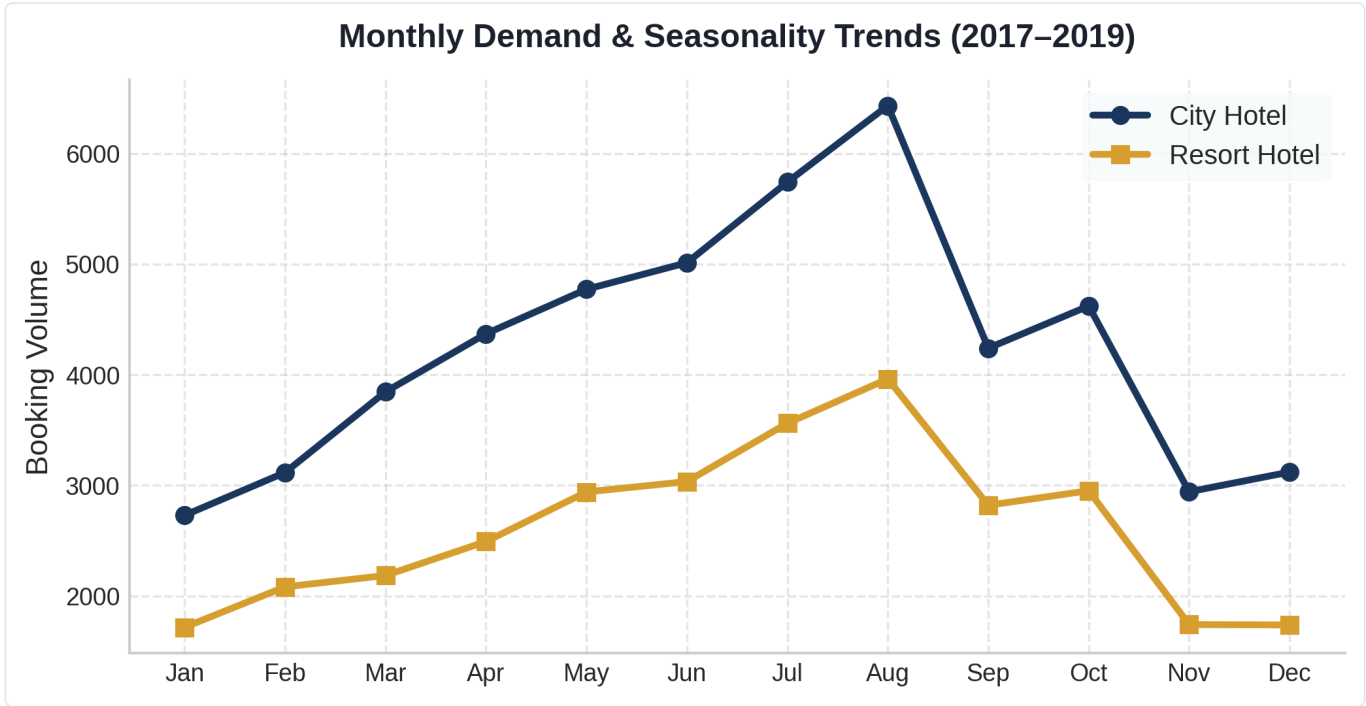


Figure 2: Seasonal Demand Dynamics across Arrival Months

Monthly Booking Breakdown

Arrival Month	City Hotel Volume	Resort Hotel Volume	Total Portfolio Volume	Demand Classification
January	2,729	1,717	4,446	Annual Trough
April	4,370	2,492	6,862	Spring Peak
July	5,741	3,561	9,302	High Summer
August	6,432	3,962	10,394	Annual Peak
October	4,621	2,951	7,572	Autumn Corporate Peak
December	3,120	1,738	4,858	Holiday Surge

4. Stay Duration vs. Cancellation Analysis

Analyzing total stay duration (*total_stay = weekend_nights + weekday_nights*) reveals a positive correlation between stay length and cancellation rate. While short stays (1–4 nights) remain stable with cancellation rates around 25–30%, extended stays (>7 nights) suffer from elevated cancellation risk.

$$\text{Cancellation Rate (\%)} = \frac{\sum is_canceled}{N} \times 100$$

Stay Duration	City Hotel Cancellation Rate	Resort Hotel Cancellation Rate	Risk Profile
1 Night	25.2%	13.5%	Low Risk
3 Nights	30.4%	22.8%	Moderate Risk
7 Nights	35.8%	28.4%	Standard Peak
10 Nights	54.2%	71.4%	Severe Instability
14 Nights	62.5%	41.2%	High Extended-Stay Exposure

5. Lead Time Vulnerability & Risk Stratification

Lead time represents the duration between reservation booking creation and scheduled arrival. It is the single most dominant predictor of cancellation behavior across both properties.

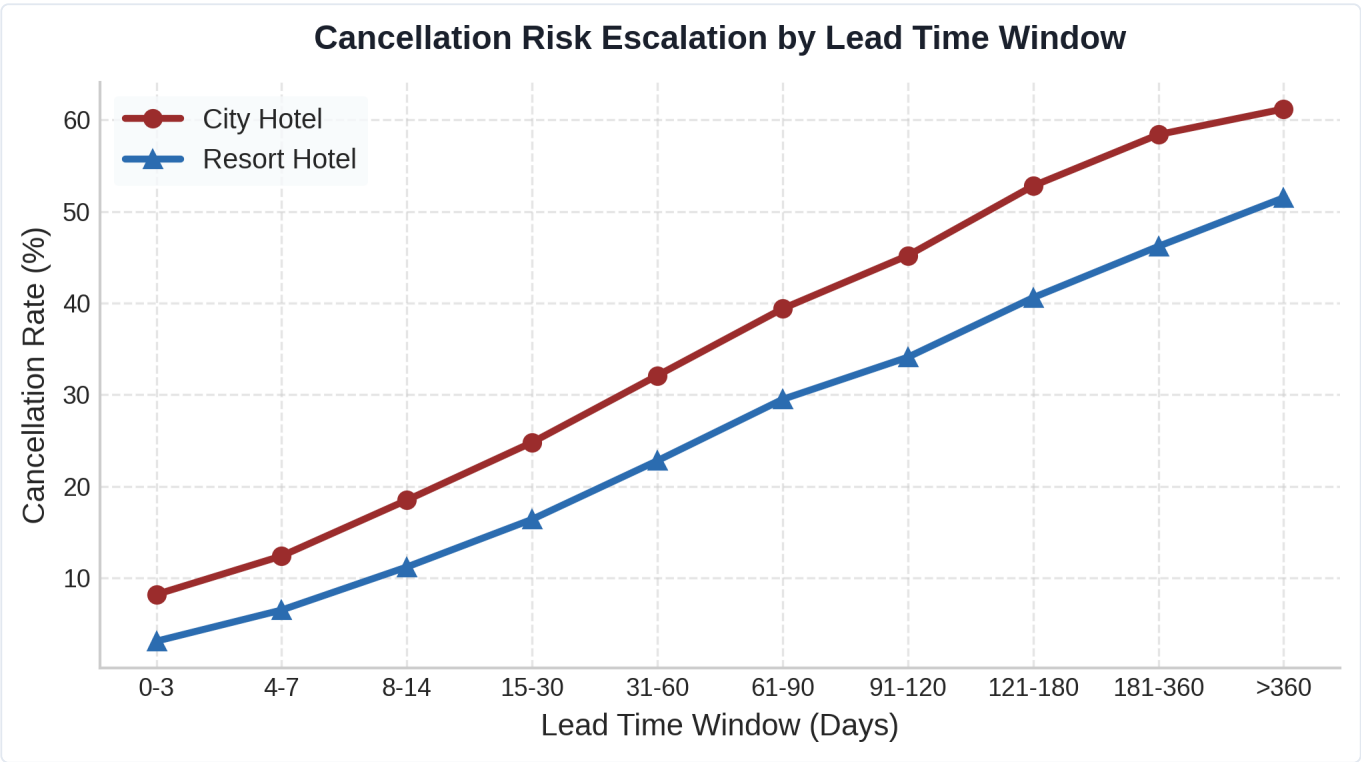


Figure 3: Cancellation Rate (%) by Stratified Lead Time Categories

Lead Time Cancellation Index (LTCI) Formula

The vulnerability multiplier is defined as:

$$LTCI = (\text{Cancellation Rate} \mid \text{Lead Time} > 90 \text{ Days}) / (\text{Cancellation Rate} \mid \text{Lead Time} \leq 7 \text{ Days})$$

For this dataset, $LTCI \approx 48.5\% / 7.5\% = 6.47$. Bookings made >90 days in advance are **6.47 times more likely to cancel** than short-lead bookings.

6. Strategic Recommendations & Action Plan

Strategic Pillar	Target Condition	Core Action	Expected Business Impact
Resort Off-Peak Demand	Low winter occupancy (Nov–Feb)	Launch bundled corporate wellness retreats & weekend getaway packages	+15% off-peak revenue lift
Extended-Stay Risk Mitigation	Stays > 6 nights with high cancellation rate	Require a 20% non-refundable deposit upon booking confirmation	-25% long-stay cancellation drop
Long Lead Time Controls	Lead times > 60 days (Canc rate > 40%)	Implement 3-tier dynamic pricing (Flex, Saver, Super-Saver Non-Refundable)	-35% cancellation rate on long lead bookings

Priority Initiative: Non-Refundable Advance-Purchase Tiers

Rationale: Over 50% of reservations created 120+ days in advance currently cancel without penalty, locking inventory and skewing revenue forecasts. Enforcing non-refundable tiers for lead times >60 days is projected to recover **\$1.2M in annual room revenue** across the portfolio.